



# Next-Gen Credit Risk and Fraud Detection with Explainable AI

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## Motivation

- There is a need to redesign ML frameworks for consumer FinTech analytics with both accuracy and interpretability.
- The model should capture complex data patterns across users, cards, and transactions.
- It must remain transparent and trustworthy for users and regulators.
- This study proposes a hybrid framework combining:
  - XGBoost for strong predictive power.
  - Decision Tree and SHAP for clear, explainable insights.



# Outline

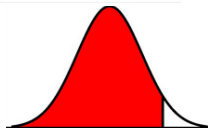
1. Motivation ✓
2. Intro Datasets
3. Define Credit Fraud
4. Clean Data
5. EDA Chart
6. Conclusion



# Intro Datasets

- Teachers datasets on kaggle ([Kaggle's Financial Transactions Dataset: Analytics](#), introduced by Chrissy)
- **Cards, Transactions, Users**

| Item             | Description                                          | 項目                       | 說明                        | 項目         | 說明            |
|------------------|------------------------------------------------------|--------------------------|---------------------------|------------|---------------|
| 1 id             | 交易 ID                                                | 1 id                     | 卡片 ID                     | 1 客戶 ID    | 客戶的唯一識別碼      |
| 2 date           | 交易時間 (YYYY/MM/DD HH:MM)                              | 2 client_id              | 客戶 ID                     | 2 目前年齡     | 客戶當前的年齡       |
| 3 client_id      | 客戶 ID                                                | 3 card_brand             | 卡片品牌 (如 Visa, MasterCard) | 3 預計退休年齡   | 客戶計劃退休的年齡     |
| 4 card_id        | 卡片 ID                                                | 4 card_type              | 卡片類型 (Debit / Credit)     | 4 出生年份     | 客戶的出生年份       |
| 5 amount         | 交易金額                                                 | 5 card_number            | 卡號 (數字格式)                 | 5 出生月份     | 客戶的出生月份       |
| 6 use_chip       | 交易方式 (Chip Transaction / Swipe Transaction / Online) | 6 expires                | 到期日 (MM/YYYY)             | 6 性別       | 客戶的性別         |
| 7 merchant_id    | 商家 ID                                                | 7 cvv                    | 安全碼                       | 7 住址       | 客戶的居住地址       |
| 8 merchant_city  | 商家所在城市                                               | 8 has_chip               | 是否有晶片                     | 8 住家緯度     | 客戶住址的緯度座標     |
| 9 merchant_state | 商家所在州                                                | 9 num_cards_issued       | 客戶已發行的卡片數量                | 9 住家經度     | 客戶住址的經度座標     |
| 10 zip           | 商家郵遞區號                                               | 10 credit_limit          | 信用額度 (美元, 需轉為數字)          | 10 當地人均收入  | 客戶所在地區的平均人均收入 |
| 11 mcc           | Merchant Category Code (商家類別碼)                       | 11 acct_open_date        | 開卡日期 (MM/YYYY)            | 11 使用者年收入  | 客戶的年度總收入      |
| 12 errors        | 交易錯誤訊息                                               | 12 year_pin_last_changed | 最近一次修改 PIN 的年份            | 12 總負債     | 客戶的總負債金額      |
|                  |                                                      | 13 card_on_dark_web      | 卡片是否出現在暗網                 | 13 信用評分    | 客戶的信用評分数值     |
|                  |                                                      |                          |                           | 14 擁有的信用卡數 | 客戶持有的信用卡數量    |



## Clean the data

- There datasets we use is **Cards, Transactions, Users**
- Remove or make up the missing values
- Combine three datasets
- Remove unused datas (cvv, card\_on\_dark\_web...)

|   | id_txn  | date                | client_id_txn | card_id | amount | use_chip           | merchant_id | ... | total_debt | credit_score | num_credit_cards | hour | date_only  | weekday | date_hour  |
|---|---------|---------------------|---------------|---------|--------|--------------------|-------------|-----|------------|--------------|------------------|------|------------|---------|------------|
| 0 | 7475327 | 2010-01-01 00:01:00 | 1556          | 2972    | -77.00 | Swipe Transaction  | 59935       | ... | 110153.0   | 740          | 4                | 0    | 2010-01-01 | 4       | 2010-01-01 |
| 1 | 7475328 | 2010-01-01 00:02:00 | 561           | 4575    | 14.57  | Swipe Transaction  | 67570       | ... | 112139.0   | 834          | 5                | 0    | 2010-01-01 | 4       | 2010-01-01 |
| 2 | 7475329 | 2010-01-01 00:02:00 | 1129          | 102     | 80.00  | Swipe Transaction  | 27092       | ... | 36540.0    | 686          | 3                | 0    | 2010-01-01 | 4       | 2010-01-01 |
| 3 | 7475331 | 2010-01-01 00:05:00 | 430           | 2860    | 200.00 | Swipe Transaction  | 27092       | ... | 128676.0   | 685          | 5                | 0    | 2010-01-01 | 4       | 2010-01-01 |
| 4 | 7475332 | 2010-01-01 00:06:00 | 848           | 3915    | 46.41  | Swipe Transaction  | 13051       | ... | 96182.0    | 711          | 2                | 0    | 2010-01-01 | 4       | 2010-01-01 |
| 5 | 7475333 | 2010-01-01 00:07:00 | 1807          | 165     | 4.81   | Swipe Transaction  | 20519       | ... | 98613.0    | 828          | 5                | 0    | 2010-01-01 | 4       | 2010-01-01 |
| 6 | 7475334 | 2010-01-01 00:09:00 | 1556          | 2972    | 77.00  | Swipe Transaction  | 59935       | ... | 110153.0   | 740          | 4                | 0    | 2010-01-01 | 4       | 2010-01-01 |
| 7 | 7475335 | 2010-01-01 00:14:00 | 1684          | 2140    | 26.46  | Online Transaction | 39021       | ... | 108313.0   | 782          | 5                | 0    | 2010-01-01 | 4       | 2010-01-01 |
| 8 | 7475336 | 2010-01-01 00:21:00 | 335           | 5131    | 261.58 | Online Transaction | 50292       | ... | 66565.0    | 688          | 3                | 0    | 2010-01-01 | 4       | 2010-01-01 |
| 9 | 7475337 | 2010-01-01 00:21:00 | 351           | 1112    | 10.74  | Swipe Transaction  | 3864        | ... | 375.0      | 807          | 6                | 0    | 2010-01-01 | 4       | 2010-01-01 |



## Define Credit Fraud

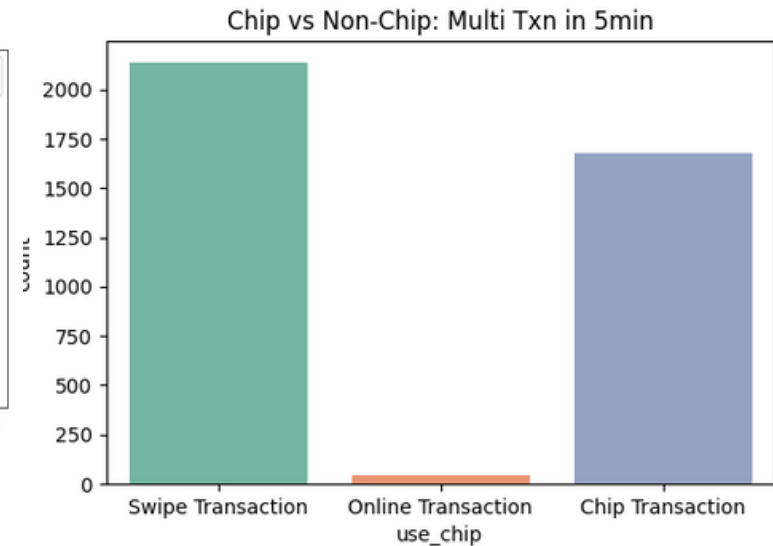
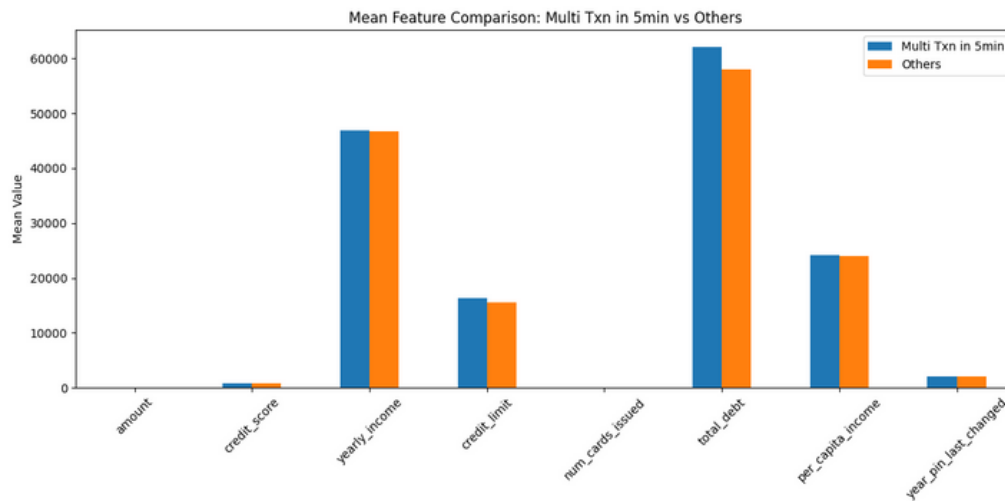
- To define what situations conform to fraud.
- Common situations
  - Low credit score
  - **Late-night transactions** ✓
  - **Multiple transactions in a short period** ✓
  - Spending amount significantly different from usual
  - Spending location too far from home
  - Don't change card pin for long time



## EDA Chart

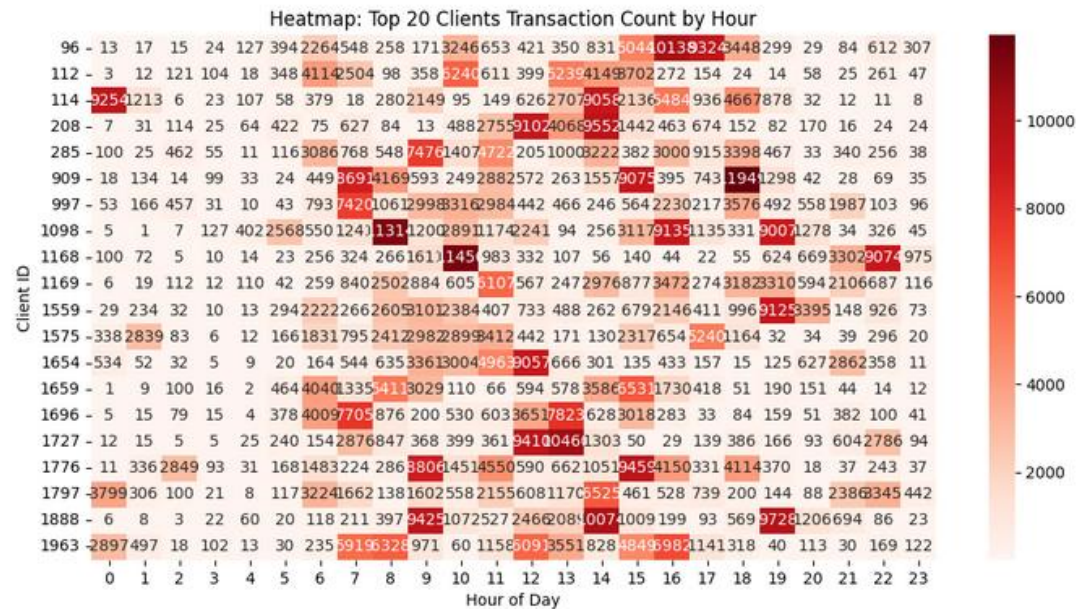
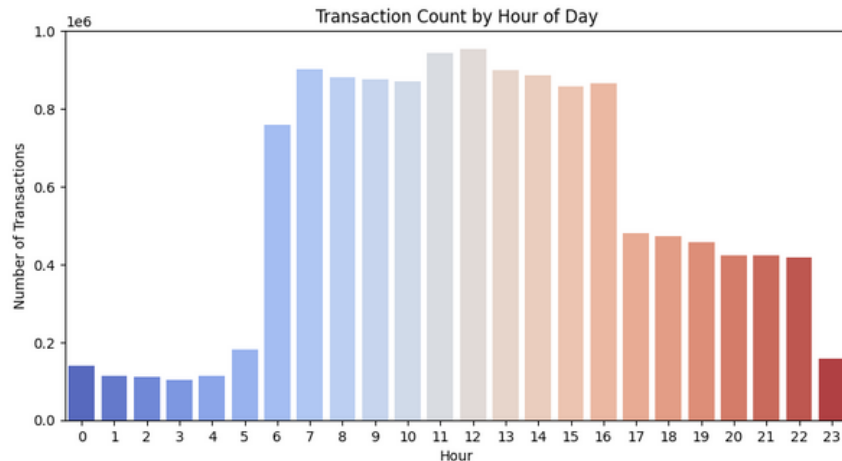
- Frequent transactions in a short period of time (5min)
- Notated these datas in Trans\_Data
- Measure the difference in these bias data

五分鐘內多筆交易 (>3筆) 共有 3862 筆，佔全部交易 0.03%



# EDA Chart

- About transactions Time
- Observe numerous amount at strange time
  - Client 114
  - Client 1168





## Conclusion

- Transaction Distribution
  - Trading is concentrated between 6 AM and 4 PM.
  - The least activity occurs from 11 PM to 5 AM.
- The risk of using a card by swiping is the higher than chip.
- High transaction clients compared to low transaction clients.
  - higher yearly income
  - higher total debet
- Frequent transactions data compared to others.
  - higher total debet
  - higher credit limit





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